

Ex. No 5 Use Autopsy to create a case and import evidence

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link:

https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

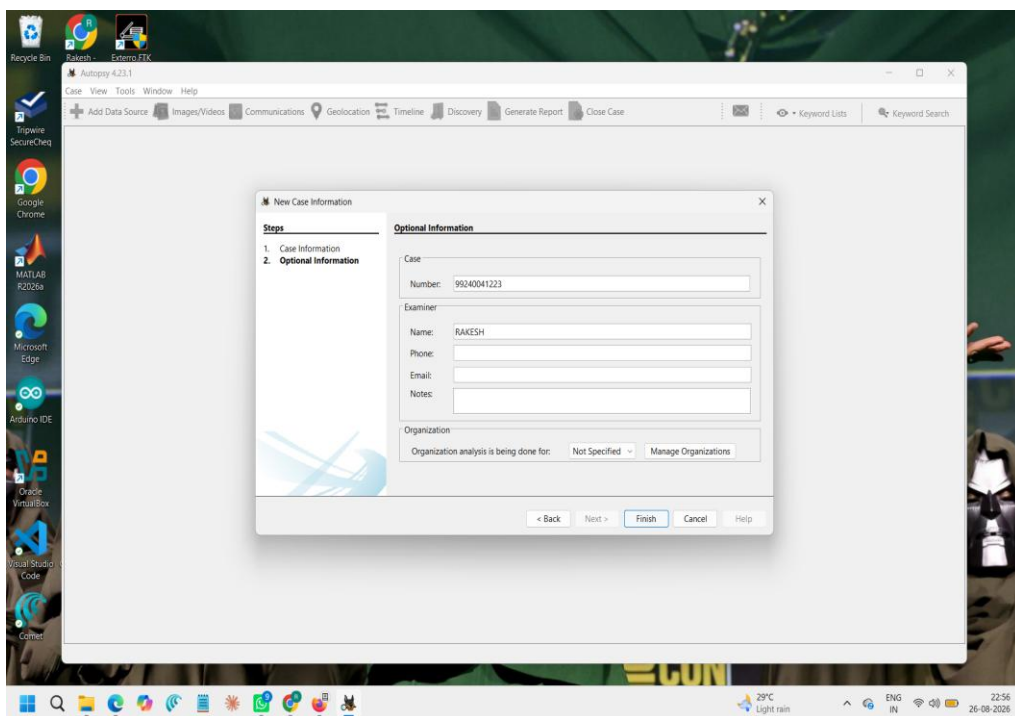
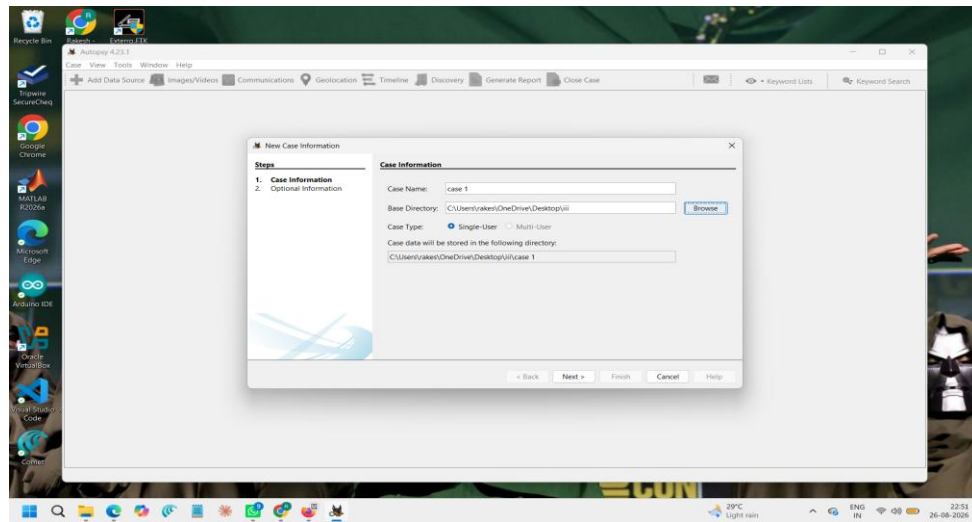
Download files: 4Dell Latitude CPi.E01, 4Dell Latitude CPi.E02

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.



3. Adding a Data Source

- **Choose the Type of Data Source:**

- After creating a case, you'll be prompted to add a data source.
- You can add disk images, directories, logical files, or local disks.**Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Local files and folders

C:\Users\thanu\Downloads\4Dell Latitude CPi.E01
C:\Users\thanu\Downloads\4Dell Latitude CPi.E02

Add
Delete
Clear

Logical File Set Display Name: Default Change

Timestamps To Include:

☐ Modified Time - Often not changed when a file is copied

☐ Creation Time - Often changed when a file is copied

☐ Access Time - Can be changed when the file is opened

NOTE: Time stamps may have changed when the files were copied to the current location.

< Back Next > Finish Cancel Help

- **Configure Ingest Modules:**
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

4. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**

The screenshot displays the Autopsy 4.23.1 interface. On the left, the 'Tree View' shows a hierarchy of data sources and artifacts. The main pane is titled 'Listing' and shows 'By Extension' results. A table lists file types and their corresponding extensions. The bottom status bar indicates 'Analyzing files from 4Dell Latitude CPl.E01' with a progress bar at 34%.

File Type	File Extensions
Images (652)	.jpg, .jpeg, .png, .psd, .nef, .tiff, .bmp, .tec, .tif, .webp
Videos (34)	.aaf, .3gp, .asf, .avi, .m1v, .m2v, .m4v, .mp4, .mov, .mpeg, .mpg, .mpe, .mp4, .m, .wmv, .mpv, .flv, .swf
Audio (146)	.aiff, .aif, .flac, .wav, .m4a, .ape, .wma, .mp2, .mp1, .mp3, .aac, .mp4, .m4p, .m1a, .m2a, .m4r, .mpa, .m3u, .mid, .midi, .ogg
Archives (162)	.zip, .rar, .7zip, .7z, .arj, .tar, .gzip, .bzip, .bzip2, .cab, .jar, .cpio, .ar, .gz, .tgz, .bz2
Databases (6)	.db, .db3, .sqlite, .sqlite3
Documents	'.htm', '.html', '.doc', '.docx', '.odt', '.xls', '.xlsx', '.ppt', '.pptx', '.pdf', '.txt', '.rtf'
Executable	'.exe', '.msi', '.cmd', '.com', '.bat', '.reg', '.scr', '.dll', '.ini'

Analysing files from 4Dell Latitude CPl.E01 34% (3 more...)

- You can perform specific keyword searches using the Keyword Search module.
- Use pre-configured lists or enter custom keywords.
- **File Analysis:**
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**
 - Compare file hashes with known databases to identify known good or bad files.

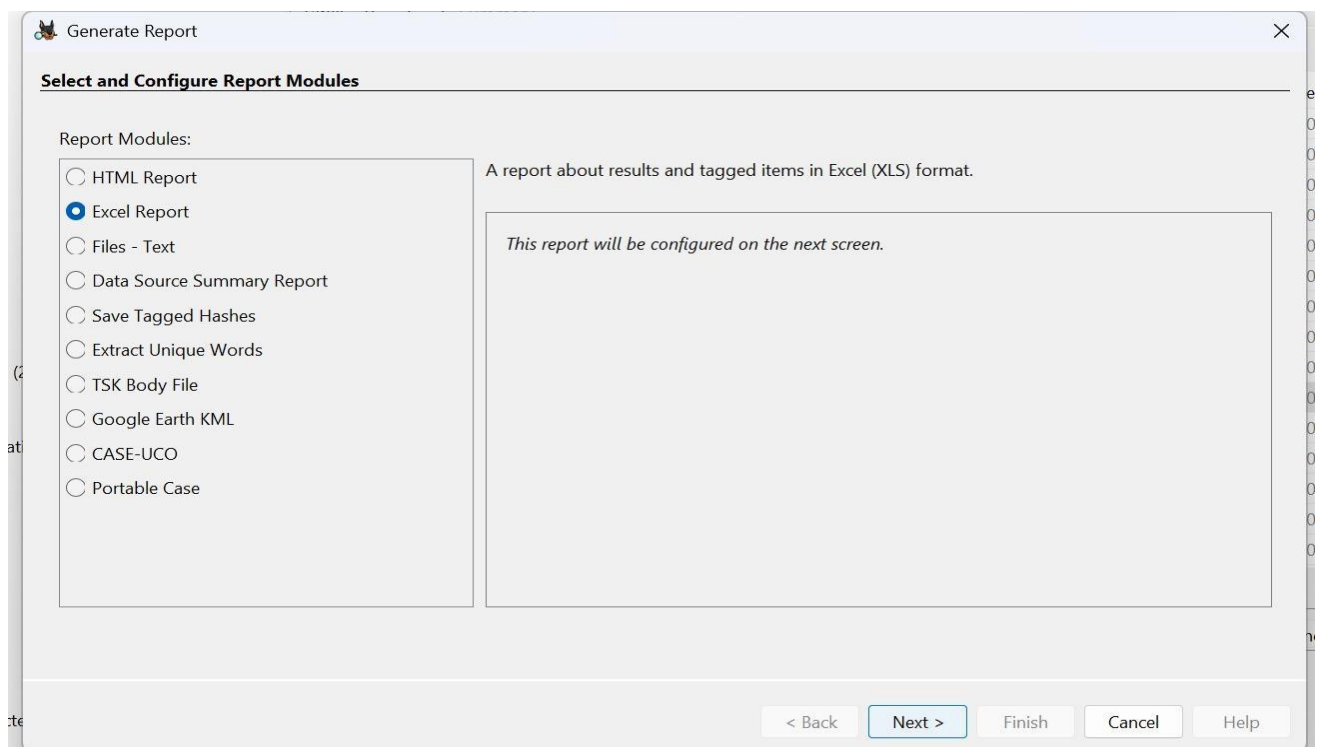
6. Reporting

- **Generate a Report:**

The screenshot displays the Autopsy 4.23.1 interface. The left sidebar shows the 'Data Sources' tree with 'Web History (887)' selected. The main pane shows a 'Listing' view of the 'Web History' table, displaying 887 results. The table has columns for Source Name, S, C, O, URL, Date Accessed, Program Name, and Domain. The data shows various web visits from 2004-08-20 to 2004-08-27, primarily using Internet Explorer.

Source Name	S	C	O	URL	Date Accessed	Program Name	Domain
index.dat			1	atwola.com/	2004-08-20 19:05:19 IST	Internet Explorer Analyzer	atwola.com
index.dat			1	cnn.com/	2004-08-20 19:05:21 IST	Internet Explorer Analyzer	cnn.com
index.dat			1	search.msn.com/	2004-08-20 21:27:41 IST	Internet Explorer Analyzer	msn.com
index.dat			1	yahoo.com/	2004-08-25 15:26:44 IST	Internet Explorer Analyzer	yahoo.com
index.dat			1	revenue.net/	2004-08-20 21:21:55 IST	Internet Explorer Analyzer	revenue.net
index.dat			1	www.cnn.com/	2004-08-20 19:05:19 IST	Internet Explorer Analyzer	cnn.com
index.dat			1	fastclick.net/	2004-08-25 15:51:06 IST	Internet Explorer Analyzer	fastclick.net
index.dat			1	doubleclick.net/	2004-08-25 15:25:45 IST	Internet Explorer Analyzer	doubleclick.net
index.dat			1	atdmt.com/	2004-08-27 15:42:53 IST	Internet Explorer Analyzer	atdmt.com
index.dat			1	servedby.advertising.com/	2004-08-25 15:51:11 IST	Internet Explorer Analyzer	advertising.com
index.dat			1	centrport.net/	2004-08-25 15:49:41 IST	Internet Explorer Analyzer	centrport.net
index.dat			1	list.ru/	2004-08-25 15:54:05 IST	Internet Explorer Analyzer	list.ru
index.dat			1	advertising.com/	2004-08-25 15:51:11 IST	Internet Explorer Analyzer	advertising.com
index.dat			1	tribalfusion.com/	2004-08-25 15:51:06 IST	Internet Explorer Analyzer	tribalfusion.com
index.dat			1	www.maktoob.com/MAIN/	2004-08-27 15:44:35 IST	Internet Explorer Analyzer	maktoob.com

- After analyzing the data, click on Generate Report from the toolbar.
- Choose the type of report (HTML, CSV, Excel, etc.).
- Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.



7. Case Closure

- **Close the Case:**

- Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**
 - Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.s

